

VINAY KUMAR

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LinkedIn GitHub DSA: [GeeksforGeeks](#) [LeetCode](#) [Portfolio](#)

Education

Delhi Technological University

2024 – 2028

B.Tech in Electrical Engineering

CGPA: 9.06

- Relevant Coursework: Data Structures & Algorithms, Circuit Theory, Analog & Digital Electronics, Control Systems, Power Systems, Electrical Machines, MATLAB

IIT Madras

2024 – 2028

BS in Data Science and Machine Learning

CGPA: 7.49

- Completed Diploma Level; Coursework: Python, Machine Learning Foundations, DBMS, Statistics, Java

Experience

Locomotive Engineering Intern

Jun 2025 – Jul 2025

Indian Railways, TKD Electric Loco Shed, New Delhi

- Worked with electrical maintenance teams on traction motors, braking systems, and locomotive power subsystems.
- Assisted engineers during **KAVACH** TCAS testing for railway anti-collision safety deployment.
- Studied relay logic, contactors, converters, and fault diagnosis in industrial railway systems.
- Supported preventive maintenance activities to improve locomotive reliability and uptime.
- Observed scheduled overhauling procedures for electrical locomotive subsystems and testing units.
- Learned industrial safety standards, maintenance workflow, and fault reporting practices.

Projects

SehatSaathi | [GitHub](#) | *Tools used: Flutter, FastAPI, PostgreSQL, Gemini API*

Ongoing

- Developing an **AI-powered healthcare platform** integrating health tracking, diet planning, and intelligent assistance.
- Engineered backend APIs for user profiles, health metrics, alerts, and recommendations.
- Implemented AI-based symptom analysis and chatbot support using LLM APIs.
- Designed personalized diet and workout engines based on user health data.
- Built family health monitoring and preventive alert modules.
- Integrated scalable database models for storing medical history and health progress analytics.

QuickCart | [GitHub](#) | *Tools used: Next.js, MongoDB, Tailwind CSS, Clerk Auth*

2025

- Developed a full-stack electronics marketplace enabling users to buy and sell gadgets.
- Built seller dashboard for product listings, inventory management, and order tracking.
- Integrated secure authentication with Clerk and dynamic MongoDB data handling.
- Designed responsive UI with Tailwind CSS for seamless user experience.
- Implemented price comparison and condition-based product filtering.
- Implemented search and category filtering for faster product discovery experience.

Metal Detector (PCB Project) | [GitHub](#) | *Tools used: Analog Circuits, PCB Design, NE555 Timer*

2025

- Designed a metal detection system using LC oscillator and electromagnetic induction principles.
- Developed custom PCB using NE555 timer based sensing circuit.
- Integrated LED and buzzer for real-time audio-visual detection feedback.
- Optimized portable low-power design using 9V battery supply.
- Tested sensing accuracy on multiple metallic objects under different distances and conditions.

Vehicle Number Plate Detection | [GitHub](#) | *Tools used: MATLAB, Image Processing Toolbox*

2025

- Developed image processing system to detect and extract vehicle number plates from images.
- Applied filtering and segmentation techniques to improve recognition accuracy.
- Used geometric region analysis for precise plate localization.
- Optimized ROI-based processing to reduce computational time.
- Enhanced edge detection and character isolation for clearer plate extraction results.

Technical Skills / Course Work

- Languages:** C, C++, Python, Java, JavaScript, HTML, CSS
- Tools/Technologies:** Linux, Git, GitHub, MySQL, MongoDB, VS Code, Figma, Photoshop
- Frameworks:** ReactJS, NextJS, NodeJS, Flask, Tailwind CSS, Pandas, NumPy
- Core Electrical:** MATLAB, PSpice, PCB Fabrication, Breadboard Prototyping, Circuit Design, DSO
- Coding Profiles:** [GeeksforGeeks](#), [LeetCode](#), [CodeChef](#)

Leadership and Extracurricular

Active Member

2024 – Present

IEEE Society, Neural AI Tech Team, DTU

- Contributed to technical events, workshops, and collaborative innovation activities.

Volunteer Member

2024 – Present

NSS, DTU

- Participated in social impact initiatives and community engagement programs.

Achievements

- Winner – Research Forge Hackathon, DTU (2024)
- Winner – PhysIQ Hackathon (2024)
- Runner-Up – BAH Hackathon 2025 (ISRO Theme)
- Gold Medal (U17) and Silver Medal (U19) – State-Level Athletics
- Completed GFG 160 Days of Problem Solving